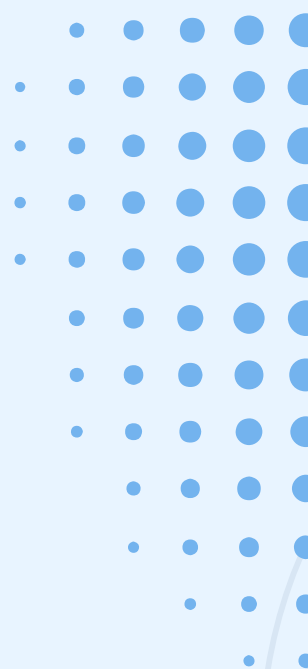
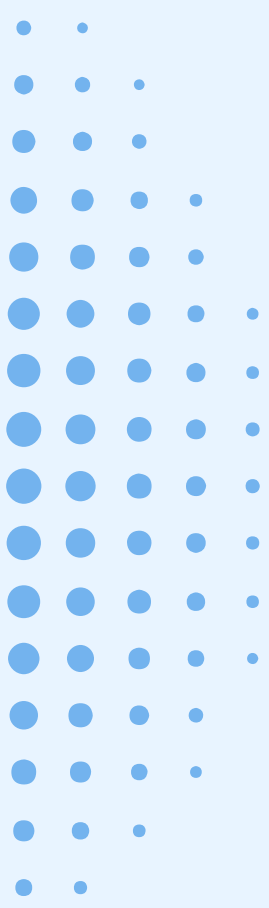




# **ANALYZING THEMATIC ALIGNMENT IN DRUG SAFETY**

Do published articles really match what a journal says it publishes?



NLP - Exam Project  
Greta Zehnder (53026A)



- EVERY JOURNAL DECLARES AN **AIMS & SCOPE**
- EDITORIAL FOCUS CAN **DRIFT OVER TIME**
- READING EVERYTHING BY HAND DOESN'T SCALE

## RESEARCH QUESTION

To what extent are the articles published in *Drug Safety* thematically aligned with the journal's stated Aims & Scope, and how has this alignment evolved between 2015 and 2025?

# THE DATASET

**DRUG SAFETY**

Scientific journal

**PUBMED**

Domain

**1,074**

Valid articles

**2015–2025**

Year range

## EDA & PREPROCESSING:

- **Year 2026**
- **“Duplicate” title**
- **Correction notices**
- **Text preprocessing**



Only 39 articles vs. 89–115 for a complete year → **excluded** as incomplete

Investigated: two distinct correction notices, **not a real duplicate**

16 short **“Correction to:”** notices **removed** (not real research abstracts)

Whitespace **normalized**; wording and casing **preserved**

# TURNING TEXT INTO VECTORS

## 01 CHUNKING

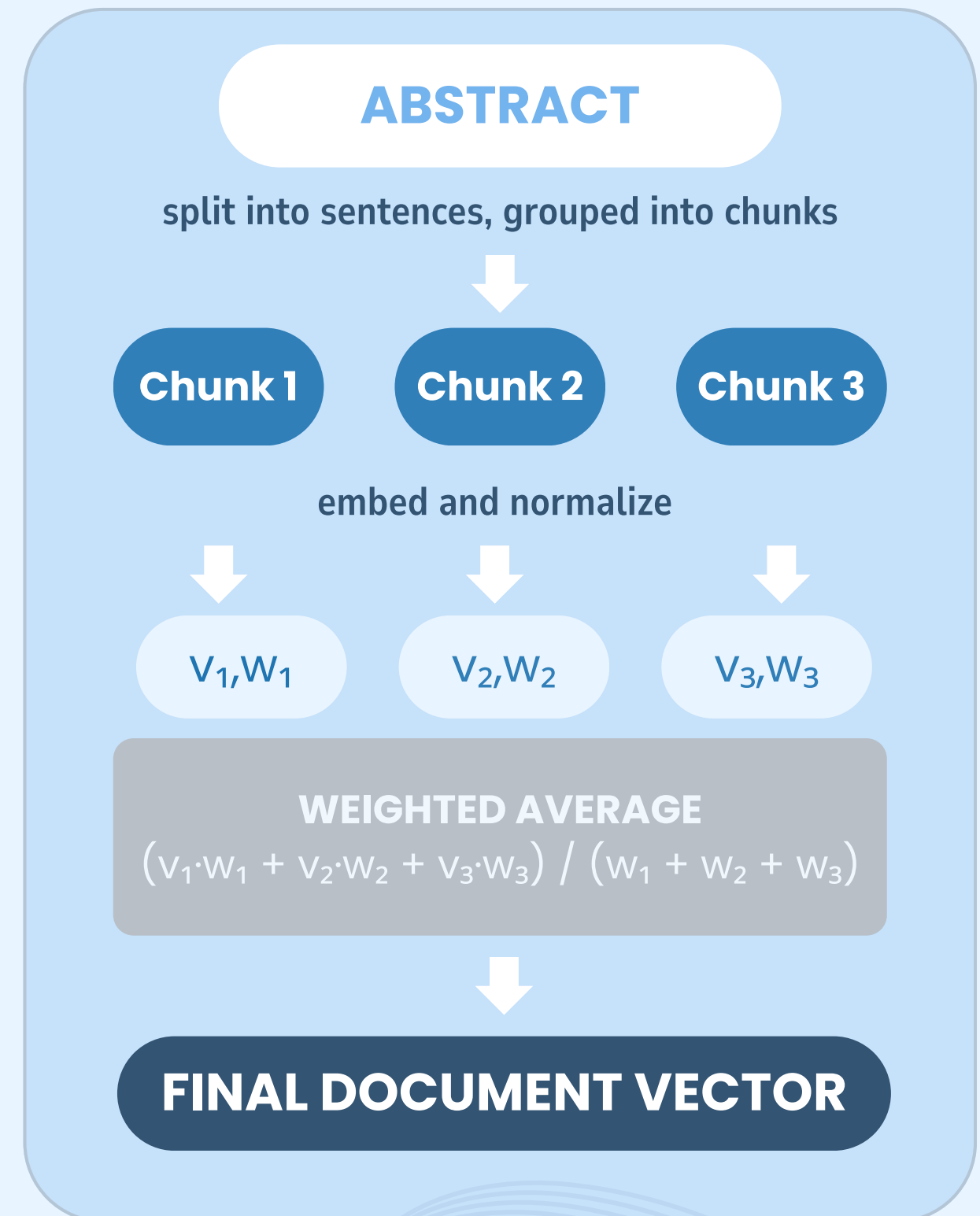
The problem: **abstracts are too long**

**92.3%** of the final corpus needed more than 1 chunk

Our solution: **sentence-based chunking**

## 02 EMBEDDING

**SentenceTransformer:** all-MiniLM-L6-v2



*w<sub>i</sub> = number of tokens in chunk i*

# ALIGNMENT SCORE AND OUTLIERS

## Alignment score

$$\text{Alignment}(i) = \cos(\mathbf{v}_i, \mathbf{v}_s) = \frac{\mathbf{v}_i^\top \mathbf{v}_s}{\|\mathbf{v}_i\| \|\mathbf{v}_s\|}$$

*cosine similarity between each abstract's vector and the Aims & Scope vector*

## Outlier detection: z-score, threshold 2.0

$$\left| \frac{\text{Alignment}(i) - \mu}{\sigma} \right| > 2.0$$

**49 outliers found:**

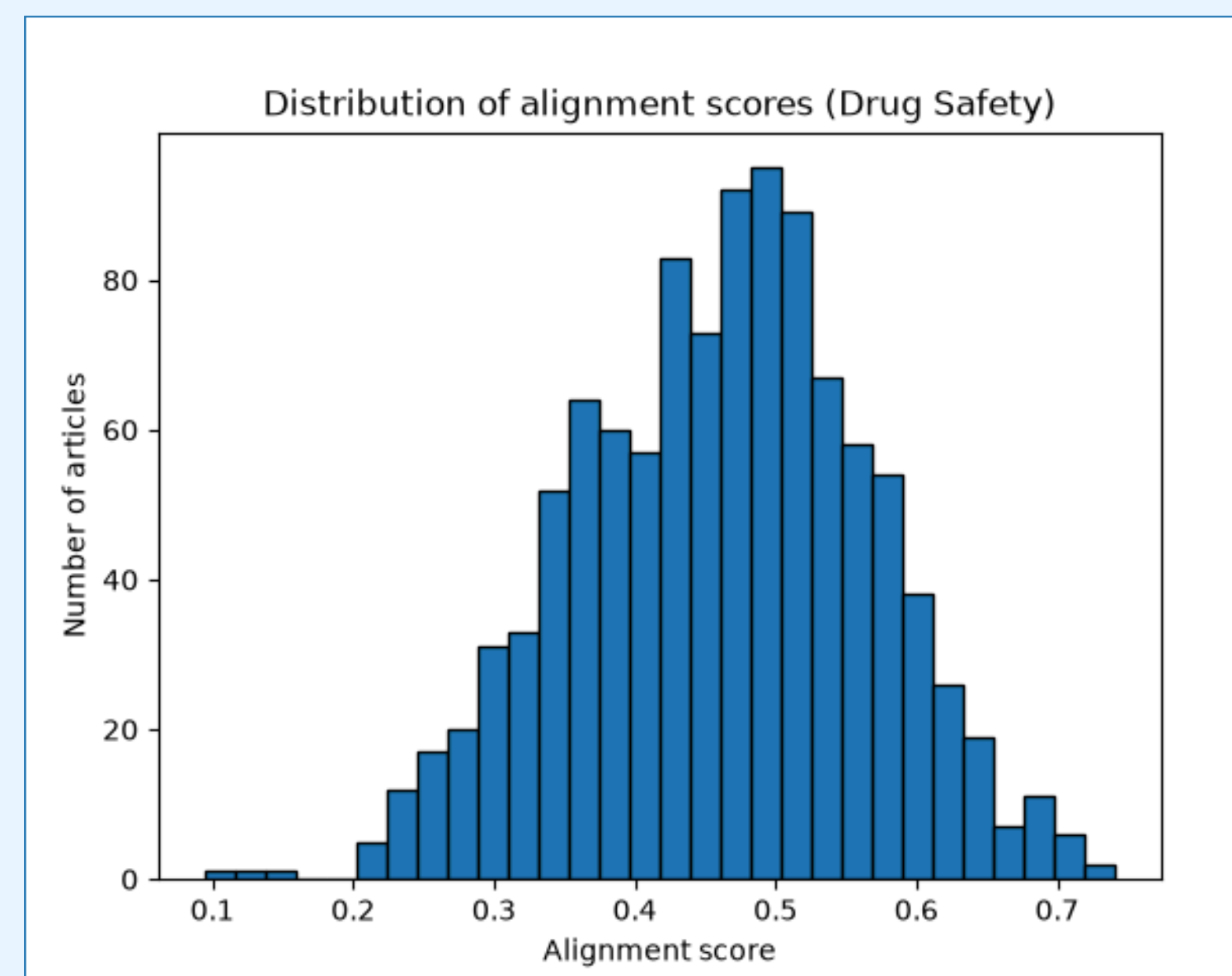
- Low-alignment outliers: 28
- High-alignment outliers: 21

**0.461**

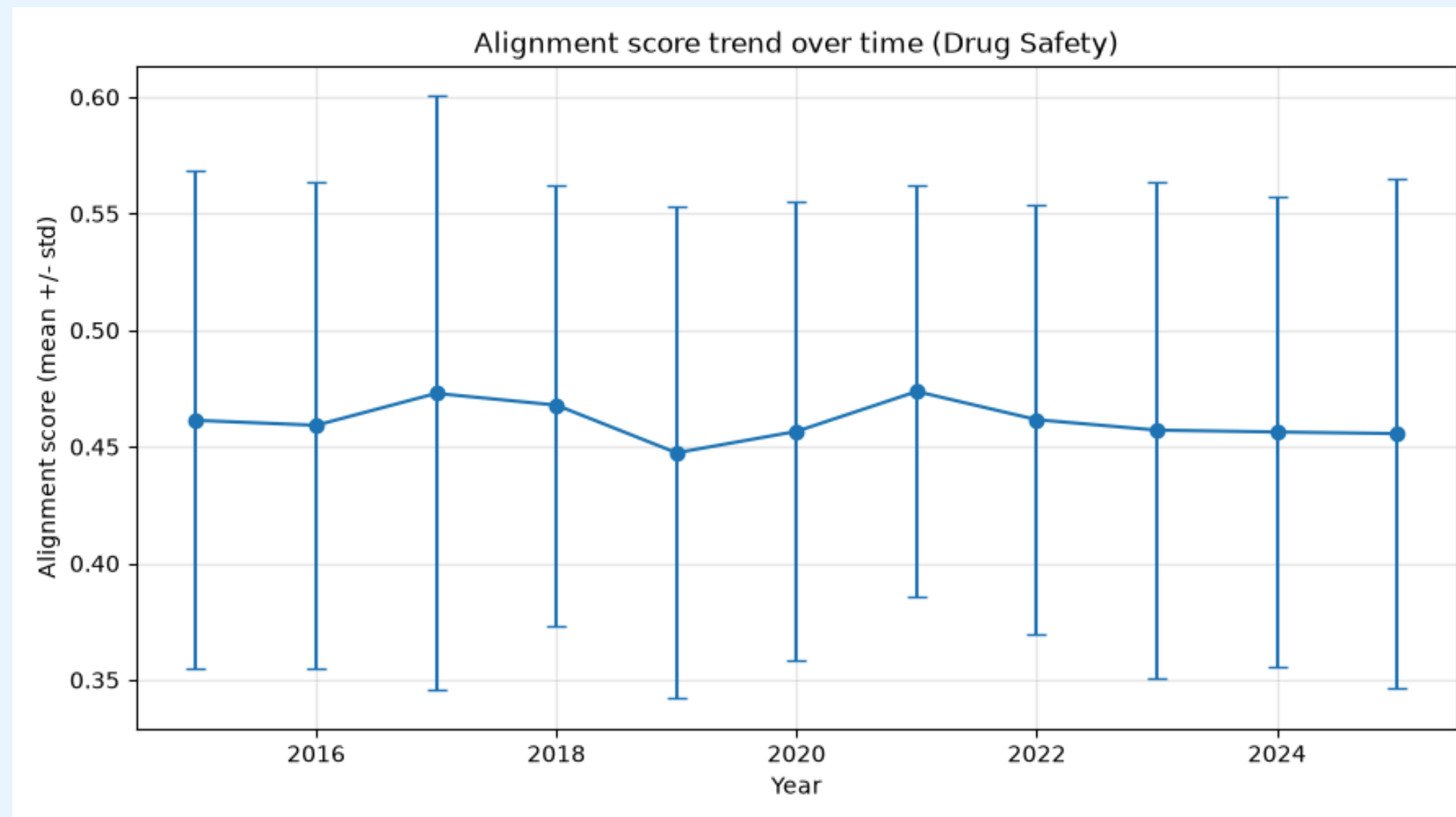
Mean score

**0.104**

Std. dev.



# ALIGNMENT OVER TIME



## A stable picture, on average

No systematic drift in the yearly average: stays within 0.45–0.47 across the whole decade.

But does that mean nothing changed underneath?

# MODEL CHOICE VALIDATION

## *Why Sentence-BERT and not raw BERT?*

Sentence-BERT provides more discriminative similarity scores

### **Sentence-BERT**

Mean 0.461

SD 0.100

Semantic-similarity optimized

### **Naive BERT**

Mean 0.841

SD 0.025

Raw mean pooling

Correlation:  $r=0.339$ ,  $p<0.001$

Naive BERT compresses scores into a narrow high-similarity range,  
while Sentence-BERT better separates articles by semantic alignment.

# READING THE EXTREMES

## HIGHEST ALIGNMENT

**0.741**

Navigating Diverging Perspectives: Reasoning, Evidence, and Decision-Making in Drug Safety

**0.727**

Summarising the Evidence for Drug Safety: Meta-Analysis Approaches

**0.716**

READUS-PV: Explanation and Elaboration

*All pharmacovigilance methodology: decision frameworks, meta-analysis, reporting standards.*

## LOWEST ALIGNMENT

**0.094**

LLM Performance & Reproducibility in Named Entity Recognition

**0.124**

Adverse Effects After HPV Vaccination — Japan

**0.158**

Narrative Search Engine with AI Query Suggestions

*Two on AI/ML methods, one on vaccine safety with narrow, specific vocabulary.*

# BERTopic

*From individual articles to thematic patterns*

BERTopic identified 24 topics + 1 noise cluster

## HIGHEST ALIGNMENT

### Topic 6

*Pharmacovigilance*  
0.577

Navigating Diverging Perspectives:  
Reasoning, Evidence, and Decision-  
Making in Drug Safety

### Topic -1

*Noise*  
0.493

Summarising the Evidence for  
Drug Safety: Meta-Analysis  
Approaches

### Topic 3

*Signal Detection*  
0.474

READUS-PV: Explanation and  
Elaboration

## LOWEST ALIGNMENT

LLM Performance &  
Reproducibility in Named Entity  
Recognition

Adverse Effects After HPV  
Vaccination — Japan

Narrative Search Engine with AI  
Query Suggestions

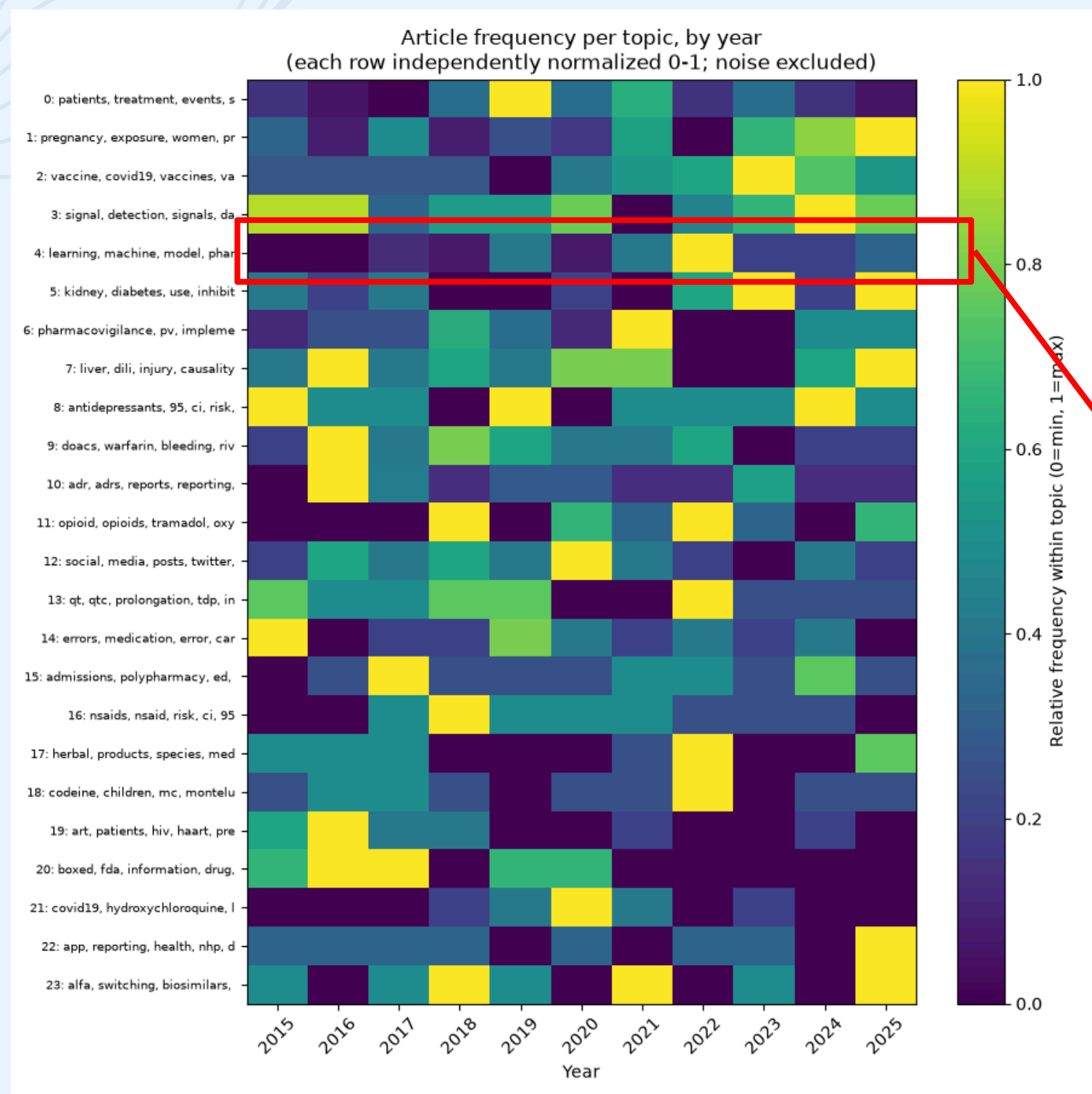
### Topic 4

*AI / Machine Learning*  
0.413

### Topic 2

*Vaccines*  
0.346

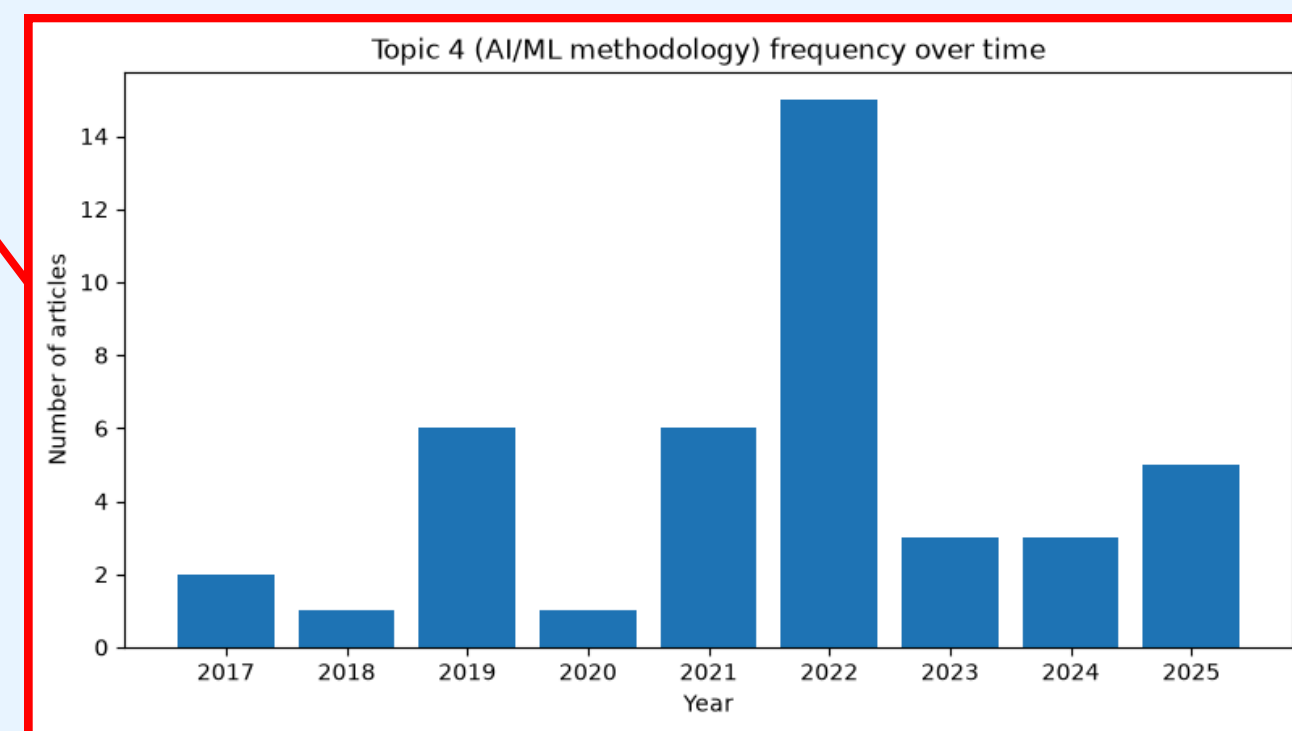
# WHAT THE FLAT TREND WAS HIDING



*Stable alignment*

$\neq$

*Stable thematic composition*



EXAMPLE: Topic 4 — AI / ML

# CONCLUSIONS

01

Overall alignment is moderate and **stable over time**

02

**Pharmacovigilance-related topics** show the strongest alignment

03

Thematic composition **changes** despite a stable aggregate score

The background is a light blue gradient. In the top-left corner, there are several thin, overlapping wavy lines in a slightly darker blue. In the top-right corner, there is a pattern of small blue dots arranged in a grid-like fashion, with some dots missing to create a sparse effect. In the bottom-left corner, there is another pattern of small blue dots, similar to the one in the top-right but with a different arrangement. In the bottom-right corner, there are more thin, overlapping wavy lines, similar to the ones in the top-left.

**THANK YOU**